



Project: Hostel Management System

CSE-347: Information System Analysis & Design

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Hostel Management System

A role-based, geo-fenced, analytics-driven web platform for modern hostel administration

Role-Based Access

Geo-Fenced Attendance

Automated Invoicing

Live Analytics

Manual Hostel Administration Doesn't Scale

Admissions, room allocation, attendance, fees, complaints, meals, visitors, and parcels — all coordinated by hand across paper files and disconnected records.



Slow & Error-Prone

Paper records delay information retrieval and invite mistakes.



Proxy Attendance

Manual roll-call is easy to fake and hard to verify.



Lost Complaints

Verbal or paper complaints are difficult to track to resolution.



Manual Invoicing

Fee tracking by hand is repetitive and error-prone.



Poor Communication

Invoice, complaint, and visitor updates don't reach students.



No Visibility

Managers lack a single view of occupancy, dues, and activity.

Problem Statement

Manual Records

Student, room, attendance and payment records are difficult to maintain and retrieve manually.

Proxy Attendance

Manual roll-call is difficult to verify and creates opportunities for proxy attendance.

Complaint Tracking

Paper or verbal complaints are difficult to monitor from submission to resolution.

Manual Invoicing

Invoice preparation and payment tracking are repetitive and error-prone.

Poor Communication

Important updates about invoices, complaints, visitors and parcels can be missed.

Limited Visibility

Managers lack a centralized view of hostel occupancy, dues and daily activities.

Required Solution

A centralized, secure web-based platform that automates hostel operations and improves efficiency, transparency and data accessibility.

Literature Review

Web-Based Hostel Systems

Existing hostel management systems commonly digitize student records, room allocation, fees, complaints and administrative workflows.

Attendance & Geofencing

Geofencing-based attendance verifies whether a user's device is inside a predefined location boundary, helping reduce proxy attendance.

Security Practices

Role-based access control and parameterized database queries are established approaches for securing multi-user web applications.

Research Gap

The proposed system integrates hostel administration, geo-fenced attendance, invoicing, notifications and analytics in one role-based platform.

[1] Awoniyi (2025) • [2] Yusof et al. (2026) • [3] Lakkimsetti et al. (2026) • [4] Geofencing Attendance (2024) • [5] Barkley et al. (1998) • [6] OWASP

OBJECTIVE

What the Project Set Out to Do

Design, develop, and deploy a centralized web platform that replaces manual hostel record-keeping end to end.



Unified web-based platform for hostel.
Role-based access for Manager, Staff, and Student.



Digitize Core Operations
Registration, room allocation, invoices, complaints, mess, visitors & attendance.



Implement GPS Attendance
Geo-fenced attendance for students and staff.



Ensure Security & Scalability
Password hashing, RBAC, parameterized queries & HTTPS.

Three Roles, One Platform

A Flask + MySQL client-server application. Each role sees only the dashboard and actions relevant to them, enforced by session-based, role-guarded routing.



Manager

- Student registration & room allocation
- Invoice generation & payment tracking
- Attendance oversight & analytics
- Complaint, violation & mess approval
- Staff & notification management



Staff

- Register visitors at front desk
- Receive & log parcels
- Record returning students
- Mark own geo-fenced attendance



Student

- View room, hostel & invoice details
- Submit complaints & feedback
- Request mess-off periods
- GPS attendance & parcel pickup

Requirement Analysis

Manager — Functional Requirements

- Manage students, staff, hostels and rooms
- Allocate rooms
- Generate invoices & update payments
- Manage attendance, complaints and violations
- Manage mess, visitors, parcels and notifications

Staff — Functional Requirements

- Register visitors
- Receive and manage parcels
- Record student returns
- Mark attendance
- View relevant attendance information

Student — Functional Requirements

- View profile, room & invoices
- Submit complaints, Request mess-off
- Mark GPS attendance
- Manage in/out records
- Track/collect parcels

Non-functional Requirements:

Operational :

- Fully Web-based application accessible across modern browsers (Desktop/Mobile).
- Cloud Deployment on Render + Aiven free-tier with SSL/TLS encryption.

Performance:

- Room availability, allocation and weekly menu: ≤ 2 seconds.
- Billing/payment status update: ≤ 3 seconds.

Security

- Role-Based Access Control (RBAC) with role-guarded routing.
- Only authorized roles can perform sensitive actions.
- Script password hashing and parameterized SQL.
- HTTPS for deployed geo-fencing..

Built on a Lean Open-Source Stack

Chosen for zero licensing cost, mature documentation, and free-tier deployability.



Backend

Python 3.10+ / Flask 3.x (Jinja2)



Database

MySQL 8.0+ / MariaDB — 18 tables, 3 triggers



Frontend

HTML5, CSS3, vanilla JavaScript



Analytics

Chart.js — donut & 14-day trend charts



Geolocation

Leaflet + OpenStreetMap (no paid API keys)



Deployment

Render (web) + Aiven (MySQL over TLS)

UML Class Diagram + Use-Case Diagram

Original diagrams from the final project report, retained for academic consistency.

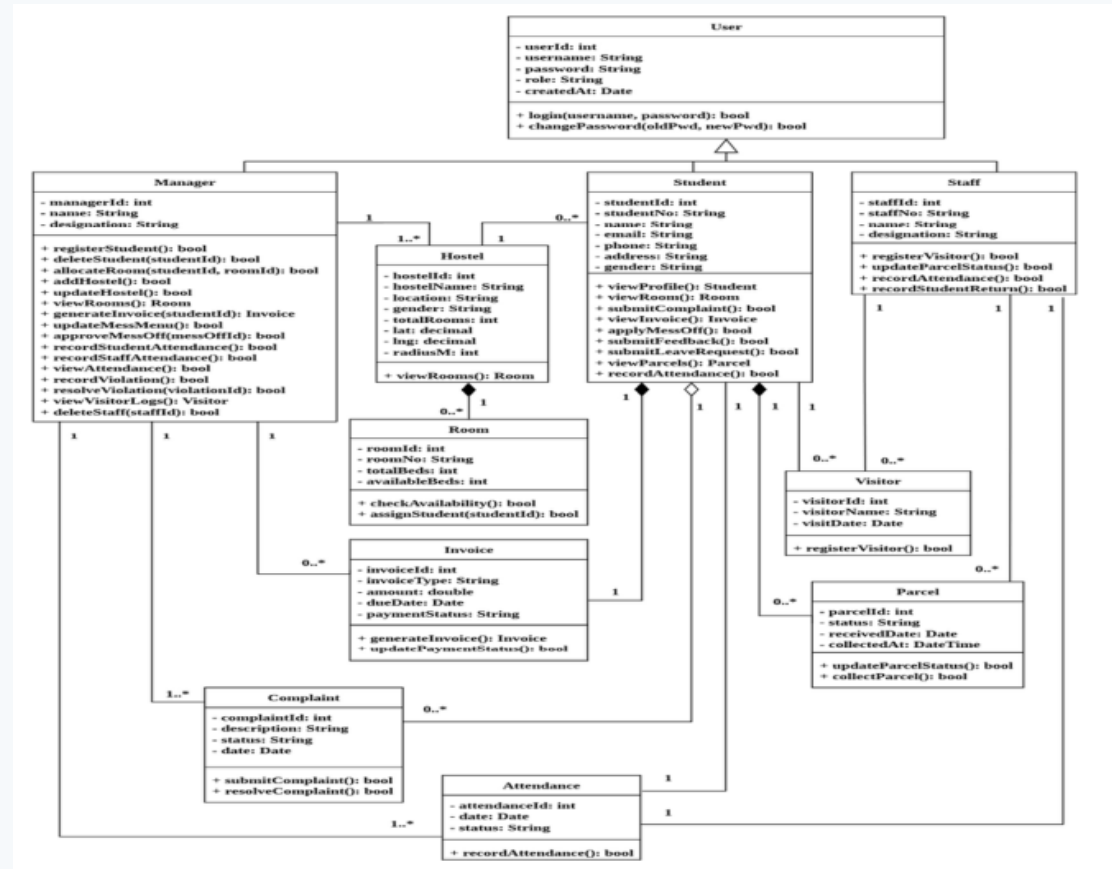


Fig-1 UML Class Diagram

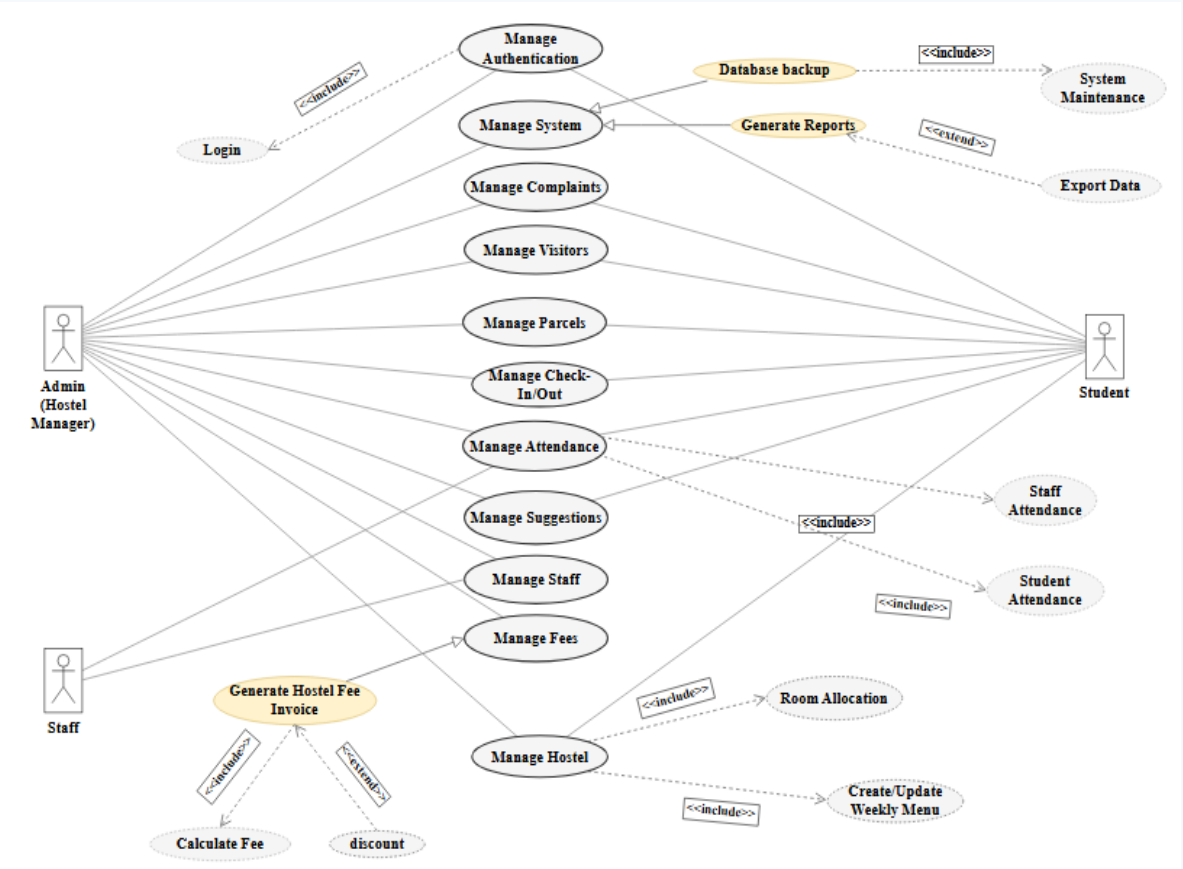


Fig-2 Use-Case Diagram

Diagrams — Sequence Flows

Examples of major system interactions

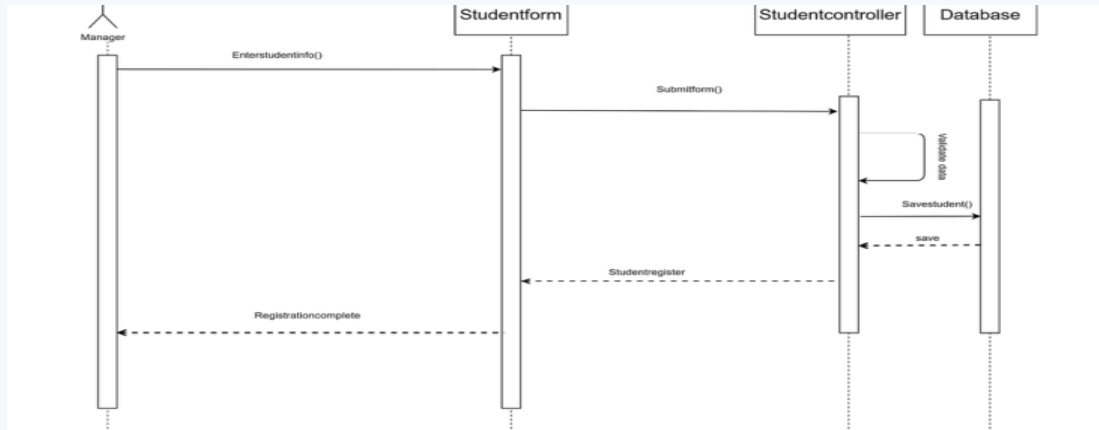


Fig-3: Sequence Diagram: Register New Students

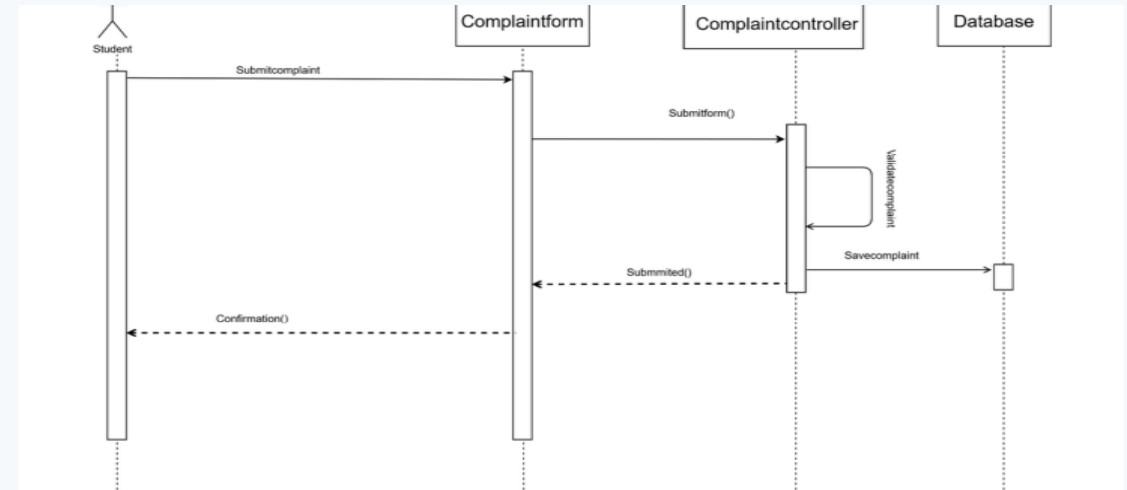


Fig-5: Sequence Diagram: Complaint

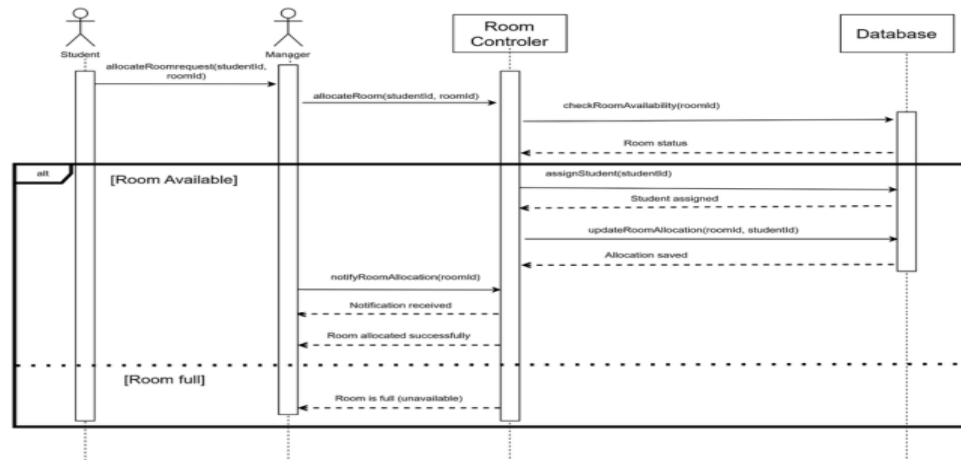


Fig-4: Sequence Diagram: Room Allocation

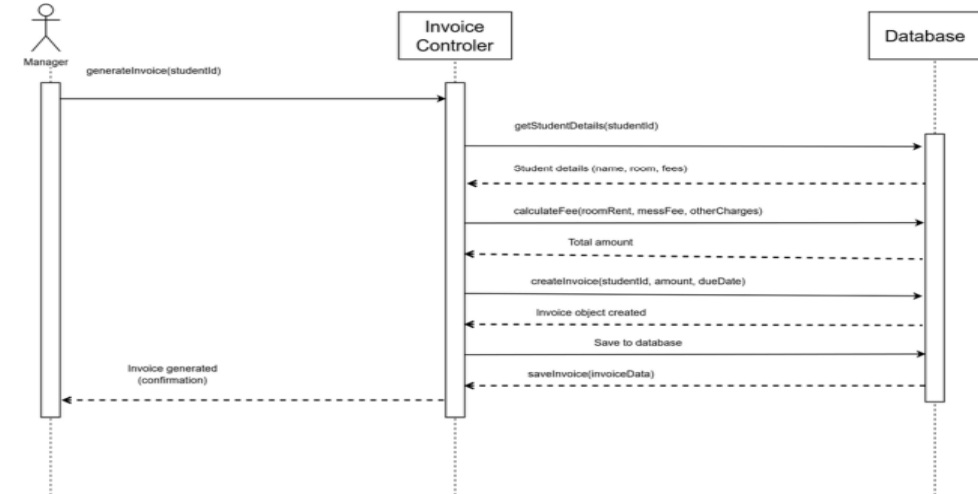


Fig-6: Sequence Diagram: Invoice

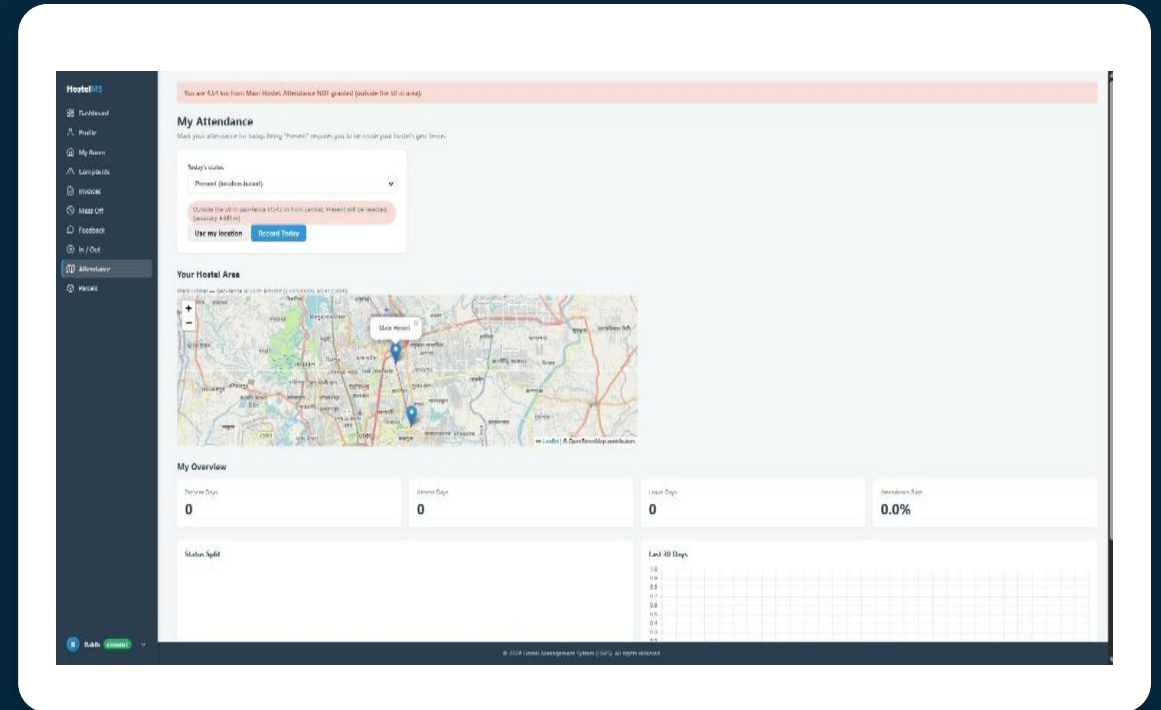
SIGNATURE FEATURE

Geo-Fenced Attendance

No proxy attendance, no dedicated hardware — just a smartphone and the browser's GPS.

- 1 Each hostel stores a latitude, longitude, and attendance radius.
- 2 The browser captures the user's live GPS coordinates on check-in.
- 3 The server computes distance with the Haversine formula.
- 4 Inside the radius → marked Present. Outside → request rejected.

Backed by nightly auto-backfill: any unmarked day is automatically recorded Absent, keeping history complete.



Live Attendance Screen — inside/outside radius check

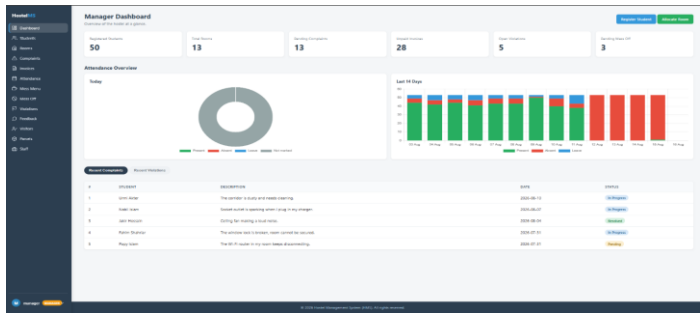
≈ 5–10 m

Typical smartphone GPS accuracy

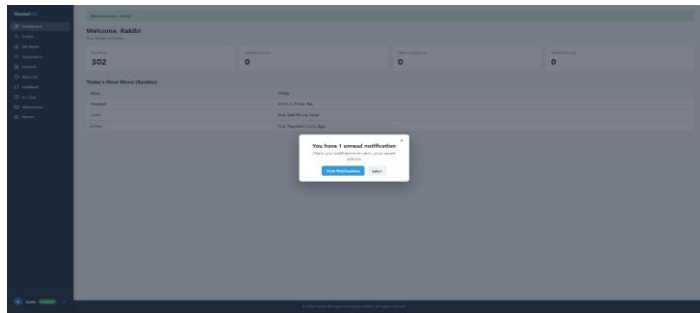
0

Extra hardware required

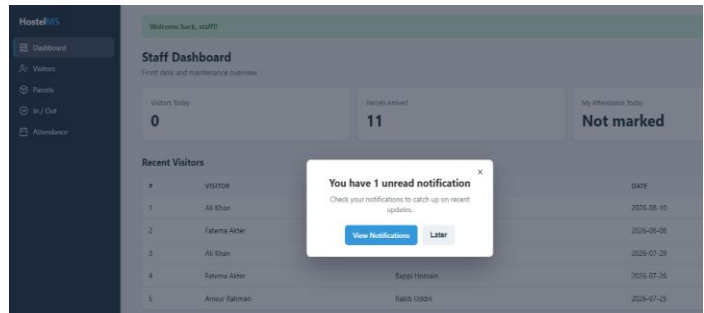
What It Looks Like in Practice



Manager Dashboard



Student Dashboard



Staff Dashboard

The Room Allocation form is used to assign rooms to students. It includes a sidebar with navigation options like Dashboard, Visitors, Mess Off, and Attendance. The main content area features a form with fields for 'Room Number', 'Room Name', and 'Room Type'. A 'Allocate Room' button is located at the bottom of the form.

Room Allocation

The Invoice Generation page displays a table of invoices generated for students. The table has columns for ID, Name, Room, Date, and Status. A sidebar with navigation options like Dashboard, Visitors, Mess Off, and Attendance is visible on the left.

ID	Name	Room	Date	Status
1	Ali Khan	Room 101	2024-08-10	Present
2	Fatima Akter	Room 102	2024-08-10	Absent
3	Ali Khan	Room 103	2024-08-10	Mess Off
4	Fatima Akter	Room 104	2024-08-10	Present
5	Amir Rahman	Room 105	2024-08-10	Absent

Invoice Generation

The Mess-Off Request form is used to request a mess off. It includes a sidebar with navigation options like Dashboard, Visitors, Mess Off, and Attendance. The main content area features a form with fields for 'Room Number', 'Room Name', and 'Room Type'. A 'Request Mess Off' button is located at the bottom of the form.

Mess-Off Request

Security-First, By Design



Script Password Hashing

Passwords are never stored in plaintext.



Role-Guarded Routing

Session-based access restricted by Manager / Staff / Student role.



Parameterized Queries

Every DB call is parameterized to prevent SQL injection.



Strict SQL Mode

Rejects invalid or malformed data at the database layer.



Server-Side Geo Validation

Attendance decisions never rely on client-side checks alone.



HTTPS / TLS

Encrypts communication end-to-end in the deployed environment.

CPM



Critical Path: T1 → T2 → T3 → T6 → T7 → T9 → T10 → T11 → T16 → T17 = 60 days

PERT Analysis



60.17 days

Expected project length



4.55

Project length variance



2.13 days

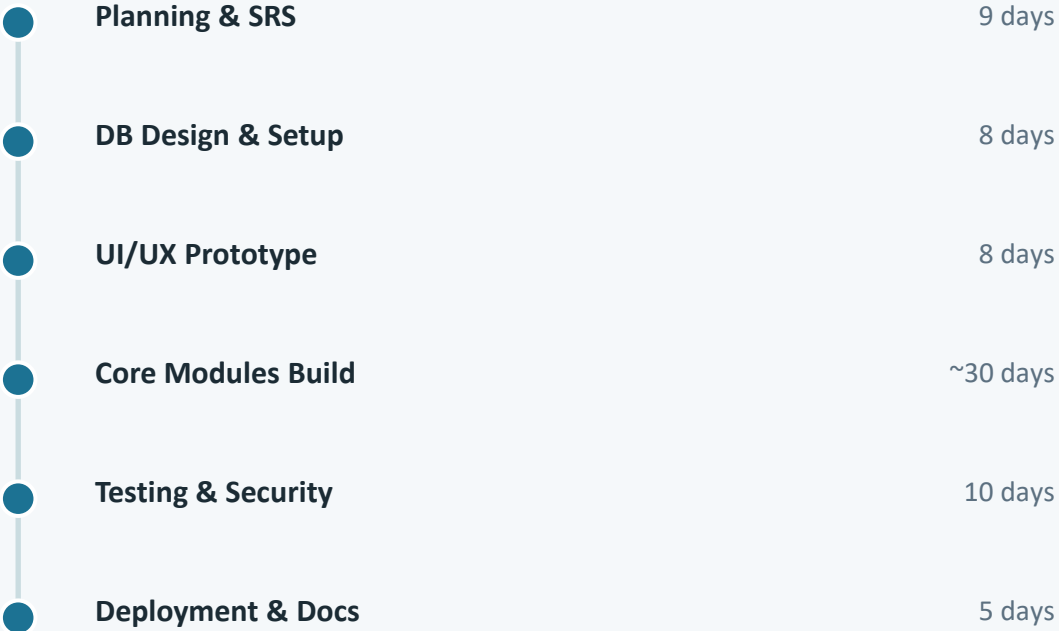
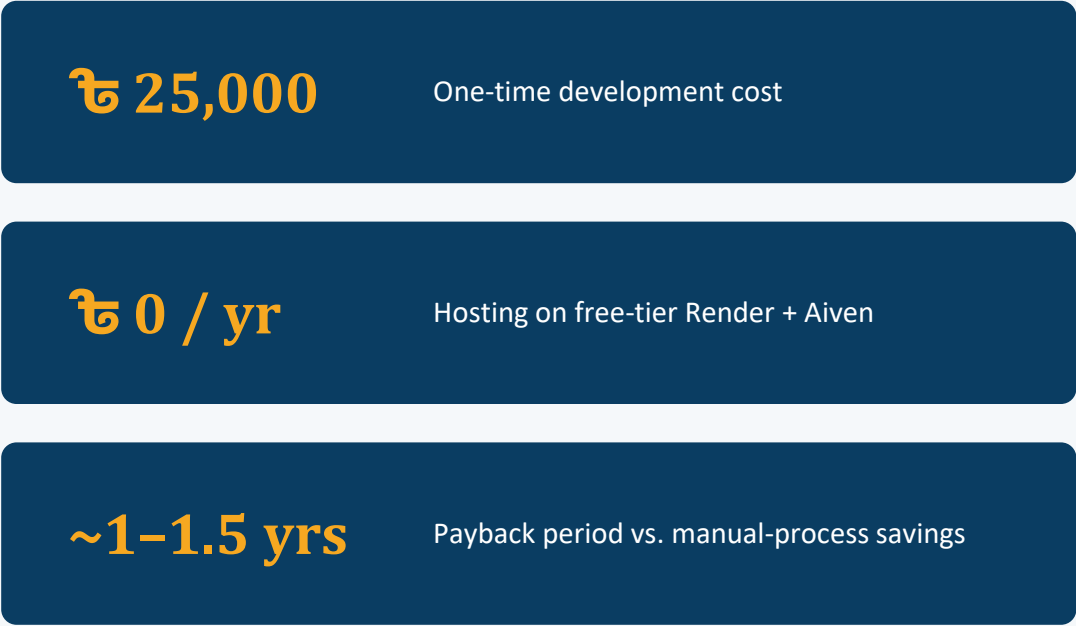
Standard deviation

PERT basis

- Uses Optimistic, Most Likely and Pessimistic estimates.
- Expected project length follows the critical activities.
- Project critical path: 1 → 2 → 3 → 4 → 7 → 5 → 8 → 10 → 11 → 13 → 16 → 17

Feasible on a Student Budget, Two-Month Timeline

Cost Snapshot (BDT)



CONCLUSION

From Paper Files to a Live Platform

The Hostel Management System replaces manual administration with a centralized, role-based web application — proving that a full-featured system can be built and deployed on open-source tools within a student budget and timeframe.

What's Next



Automated daily database backups



'Forgot Password' self-service recovery



Pagination for large attendance & invoice datasets



Printable PDF attendance report per student

18

Database Tables

3

User Roles

50

Demo Students Seeded

~60

Day Build Timeline

LIVE NOW

Deployed, Monitored, and Ready to Try

Deployment

- Render (web service) + Aiven (free MySQL over TLS)
- Same codebase runs locally (XAMPP) and on the cloud
- GET /healthz endpoint monitored via UptimeRobot
- Incremental delivery: auth → core modules → attendance/invoices → testing → deployment

Testing covered individual modules, integrated flows, all three user roles, and role-based security controls before release.



Try It Live

hostel-management-368b.onrender.com



Sign in with any demo account:

Manager

manager / admin123

Staff

staff1 / staff123

Student

student1 / student123

50 seeded students with 2.5 months of attendance history for realistic demo data.

Thank You

Questions & Discussion

Hostel Management System — CSE347